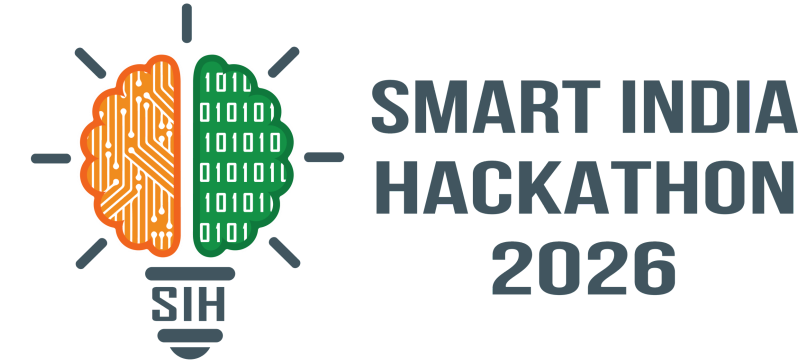


SMART INDIA HACKATHON 2026



TITLE PAGE



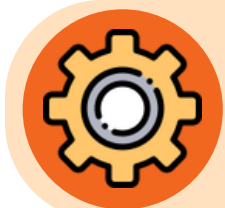
Problem Statement ID
26047



Problem Statement Title
Patient Case-Taking Software



Theme
MedTech/ Biotech/ HealthTech



PS Category
Software



Team ID
ID:40



Team Name
Code Tantra



MediKiosk is a web-based platform that **digitizes and streamlines the patient case-taking process**, using **AI-assisted analysis** to help doctors manage and review cases more efficiently.

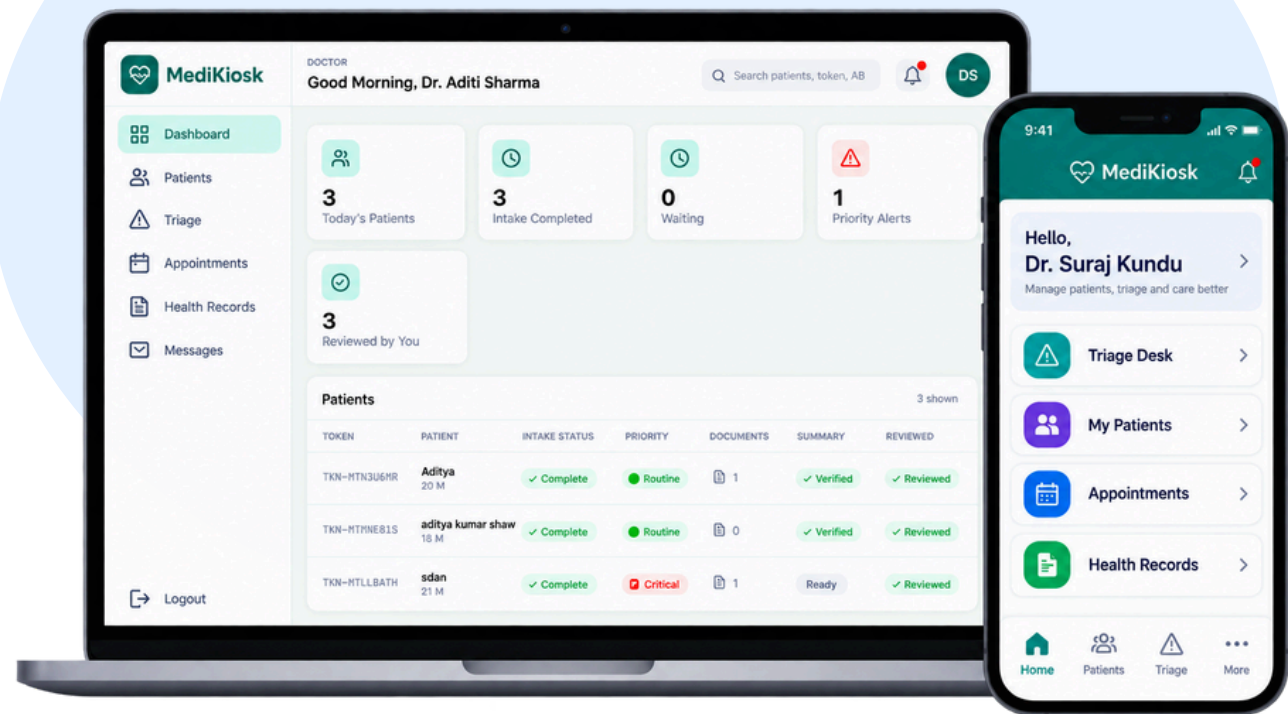
Real-World Problem

Indian government hospital OPDs see 4,000–10,000 patients/day with only 2–5 minutes per consult — leaving doctors no time for proper history-taking while patients' records stay scattered across unstructured paper documents.



Landing Page
with a Fully
Functional
Website

One platform
Multiple Users
Better Healthcare



How It Addresses the Problem

- Replace manual and paper-based case-taking with a digital and structured system.
- Reduces waiting time and repetitive explanation of symptoms.
- Helps doctors access complete patient information in one place
- Highlights urgent and high-priority cases for faster attention.
- Improves communication between patients and doctors.
- Enables better tracking and management of cases.



Innovation and Uniqueness

- Ai-assisted extraction and summarization of patient information.
- Role-based portals for Patients, Doctors and Administrators.
- Smart case prioritization (Urgent/ High/ Normal).
- Integrated document management (reports, prescriptions, scans).
- Real-time case tracking and notifications.
- Simple, user-friendly and accessible from anywhere.



Key Features


Digital Case Submission


AI-Assisted Insights


Medical Document Management

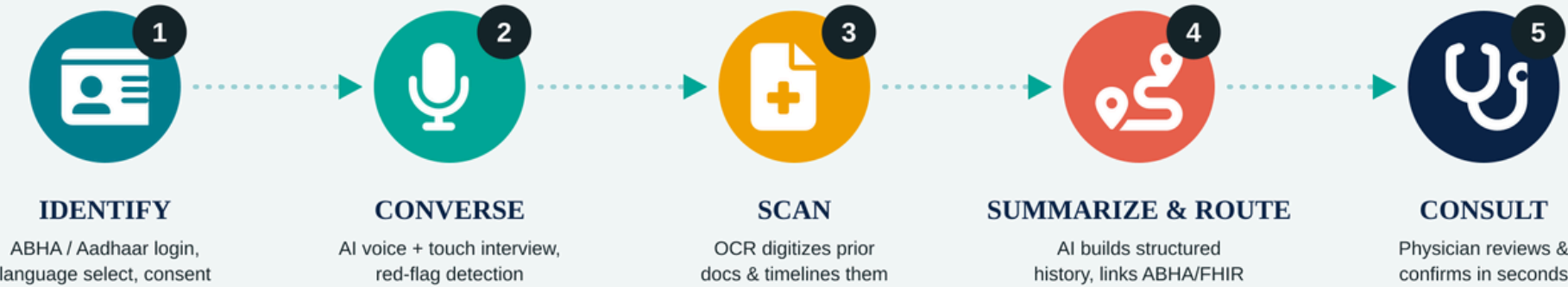

Appointment & Communication


Secure & Private

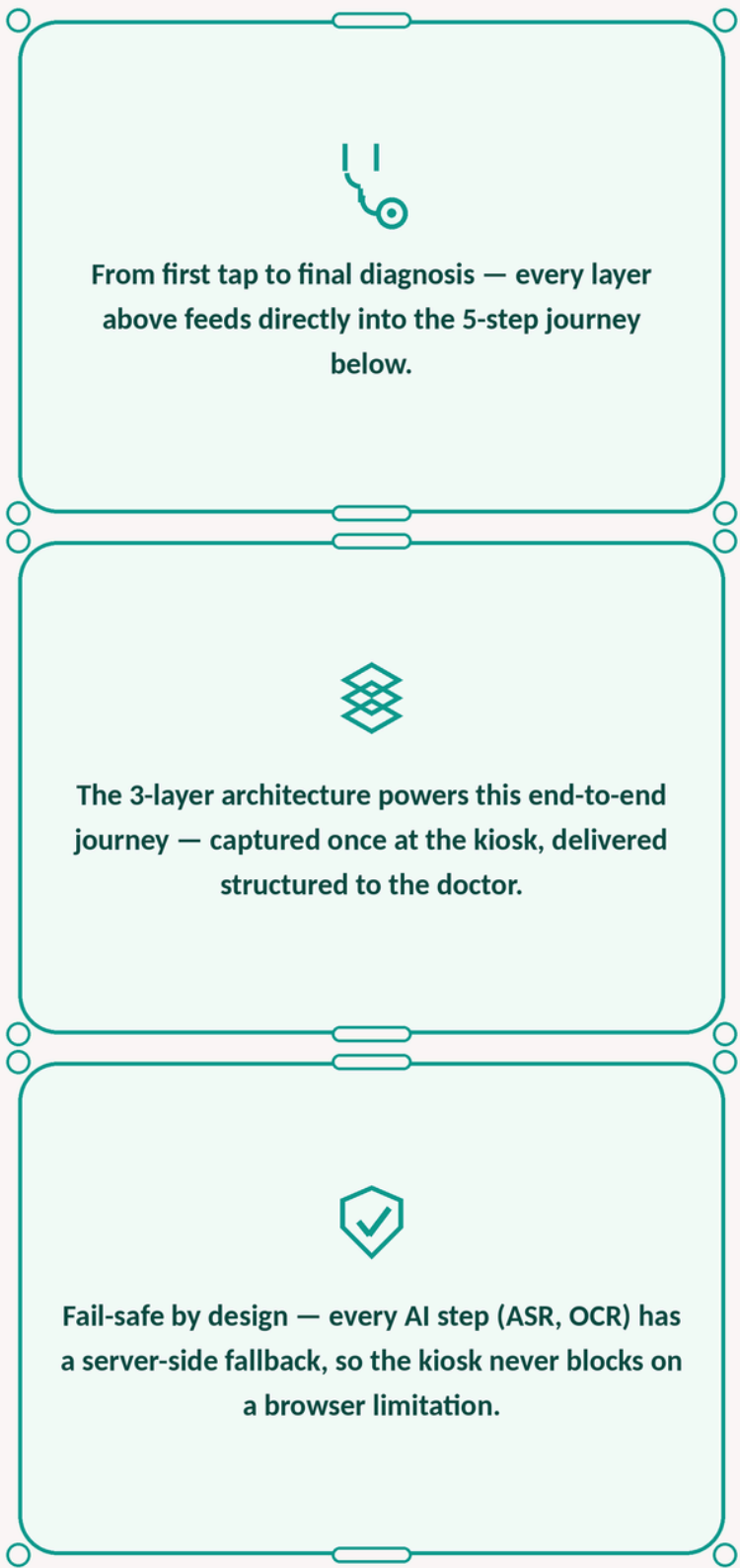
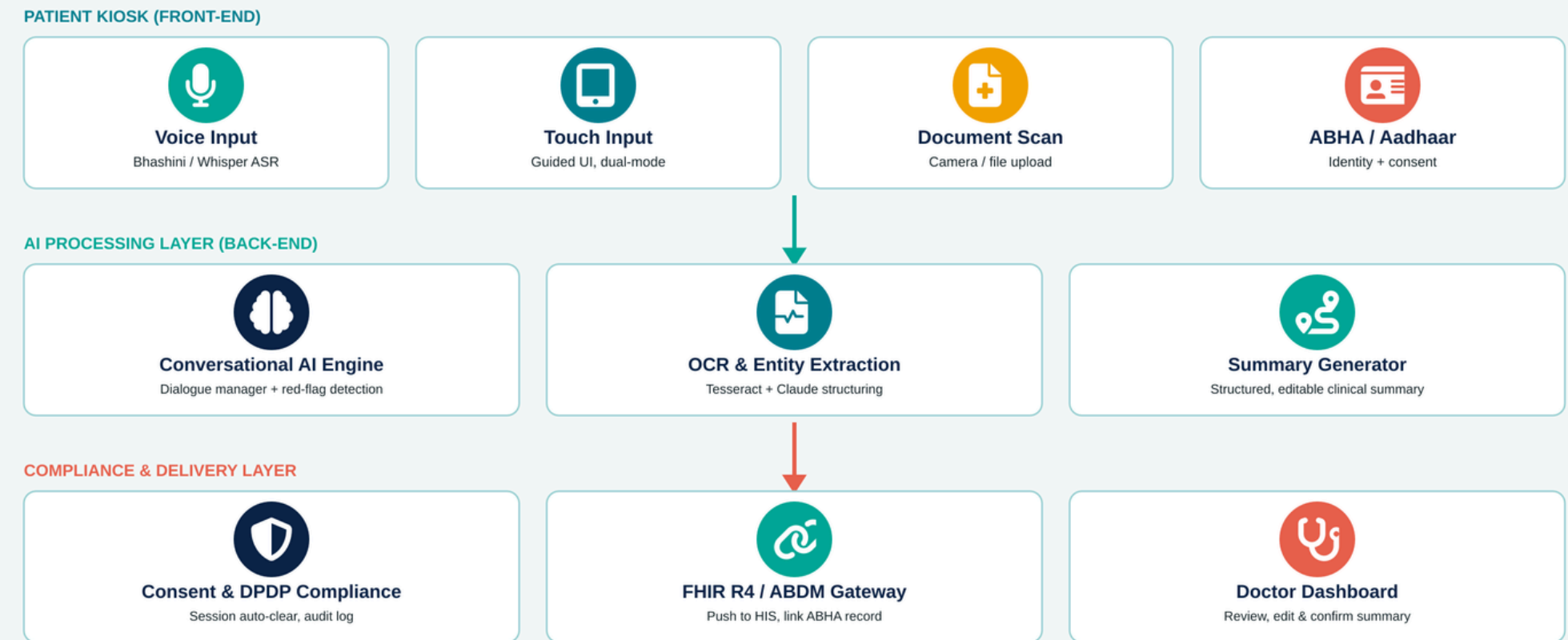

Admin Dashboard & Analytics


Scalable Hospitals and Clinics

MediKiosk — End-to-End Patient Journey



MediKiosk — System & Data Flow Architecture



MediKiosk — Technical Stack

- 1 FRONTEND**
React, Vite, Tailwind CSS — Kiosk UI, multi-language screens
- 2 BACKEND**
Node.js, Express, Mongoose — REST API and business logic
- 3 DATABASE**
MongoDB Atlas — patient records, audit log, documents
- 4 INTEGRATIONS**
ABDM, Bhashini, Tesseract OCR, Gemini API
- 5 SECURITY**
JWT auth, bcrypt, Helmet, rate-limiting — session & data safety



Feasibility Analysis

Our solution is technically, operationally and contextually feasible with existing tools and infrastructure.



Technology Feasibility

Built using proven open-source technologies (MERN stack, Tesseract.js, Whisper, FHIR, etc.).



Resource Feasibility

Can be developed and deployed by a small team with standard hardware (PC/Tablet) and open-source/affordable tools.



Operational Feasibility

Simple, kiosk-friendly user interface with multi-language support (Bhashini). Designed for use in hospitals, clinics and PHCs.



Economic Feasibility

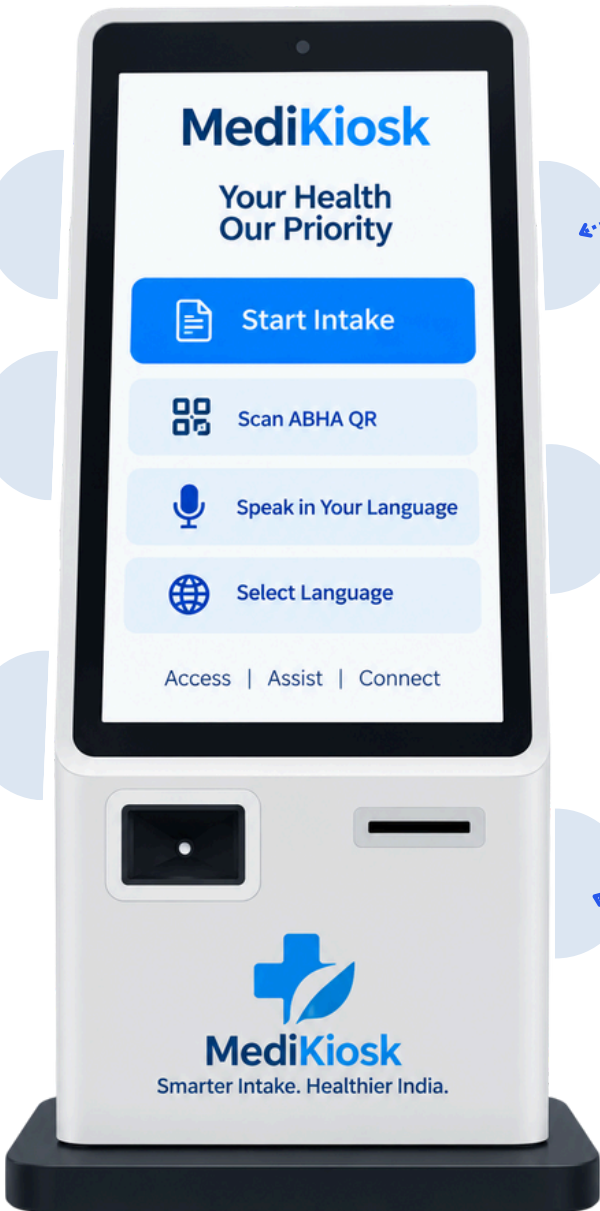
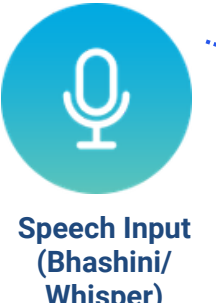
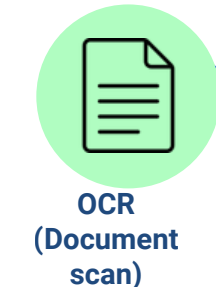
Cost-effective compared to fully manual intake. Uses open-source tools and cloud services. Can run on low-cost hardware.



Regulatory Feasibility

Aligns with India's digital health initiatives (ABDM/ABHA) using FHIR R4 standards. Designed to be compliant with NDHM guidelines.

Real Technology. Real Impact.
A practical solution for real-world healthcare settings.



Potential Challenges and Risks

We have identified key risks and are proactively planning to address them.



Data Privacy & Security

Sensitive patient data may be at risk.



Internet Dependency

Required for ABDM and Bhashini services.



User Adoption

Patients or staff may hesitate to use a new digital system



Integration with Hospital System

Different HIS systems may have varying standards.



Accuracy of AI & OCR

Errors in Document/speech interpretation may occur.



Regulatory & Compliance

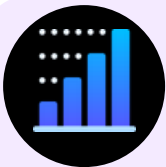
Need to ensure compliance with NDHM and data protection norms.



Strategies for Overcoming These Challenges

Turning challenges into opportunities.

- ✓ Use encryption, secure authentication and follow NDHM guidelines.
- ✓ Enable offline mode and local fallback (Whisper, simulated APIs)
- ✓ Conduct user training and design an intuitive, multi-language interface.
- ✓ Use FHIR standards and modular APIs for easier integration
- ✓ Apply human-in-the-loop review for AI/OCR outputs.
- ✓ Collaborate with healthcare stakeholders for pilots and feedback.



Overall Viability

Medikiosk is a viable, scalable and sustainable solution aligned with India's healthcare goals.



Deployable in real healthcare settings



Supports Digital India & Ayushman Bharat



Scalable from PHCs to large hospitals



High potential for long-term impact

IMPACT AND BENEFITS

Why a 5-Minute Consultation Can Now Do More

BEFORE MEDIKIOSK

- ✗ History-taking eats 3 of 4 consult minutes
- ✗ Only ~1 min left for exam + counselling
- ✗ Missed comorbidities, rushed diagnosis



WITH MEDIKIOSK

- ✓ History captured before doctor walks in
- ✓ 3.5 of 4 minutes free for exam + care
- ✓ Complete record, sharper diagnosis

6x FASTER

History-taking drops from 3.0 to 0.5 minutes per patient

70–80%

Diagnoses backed by complete history

4K–10K

OPD patients served per day

3.5 min

Extra time for exam & counselling

100%

ABDM / FHIR record coverage

Why It Matters



Sharper diagnosis

A complete history means fewer comorbidities slip through.



AYUSH gets its due time

Dashavidha Parikshya Bhava captured fully, not squeezed into 5 minutes



Emergencies flagged instantly

Red-flag symptoms escalate to triage the moment they're spoken.



One record, every visit

ABDM + FHIR keep every consultation linked and reusable.

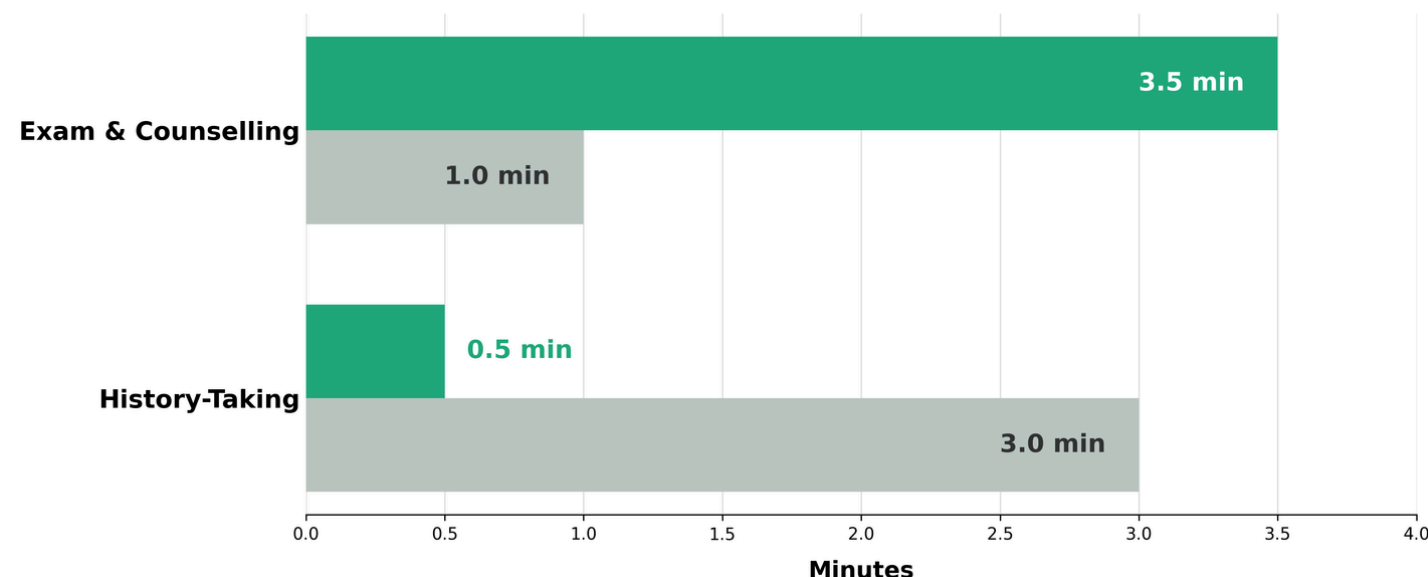


Everyone benefits

Patients get ease, doctors get time, OPDs get throughput.

Where the 4 Minutes Go

MediKiosk Current OPD



TIME REINVESTED
PER PATIENT

+2.5 MIN GAINED BACK

That's the difference between rushing a diagnosis and getting it right — redirected straight into examination and patient counselling, not paperwork.

◆ Faster Consultations

◆ Better Diagnosis

◆ AYUSH-Ready

◆ ABDM-Linked

◆ Built for Scale



Every design decision behind **MediKiosk** is backed by **clinical evidence**, **government digital-health standards**, **AYUSH literature**, and **legal frameworks** — ensuring the solution is credible and deployment-ready.



Clinical & Global Research

- **BMJ Open (2017)** — Global Consultation-Time Study
bmjopen.bmj.com/content/7/10/e017902



Government Digital Health Standards

- **Ayushman Bharat Digital Mission (ABDM)**
abdm.gov.in
- **FHIR Implementation Guide for ABDM (NRCeS/NHA)**
nrces.in/ndhm/fhir/r4/index.html
- **Bhashini — National Language Translation Mission**
bhashini.gov.in



AYUSH & Traditional Medicine

- **Dashavidha Parikshya Bhava (Charaka Samhita, PMC)**
pmc.ncbi.nlm.nih.gov/articles/PMC6891990



Legal & Technology Compliance

- **Digital Personal Data Protection Act, 2023 (MeitY)**
meity.gov.in — [DPDP Act, 2023 \(PDF\)](#)
- **Tesseract OCR — Official Repository**
github.com/tesseract-ocr/tesseract



Evidence-Based



Govt-Aligned Standards



Legally Compliant



Open-Source & Verified



[GitHub Repository](#)



[Live Demo — MediKiosk](#)